



ANALYTICAL LAB

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اسئلة سنوات

1)



Phosphoric acid Molar
Mass=98g/mol
Density=1.685g/ml
mass% =85%

- a) 0.0146M
- b) 17.2M
- c) 14.6M**
- d) 20.2M

$$M = \frac{V \times d \times w\%}{MW}$$

$$M = \frac{1000 \times 1.685 \times 85\%}{98}$$

$$\underline{\underline{M = 14.6}}$$

2)



Sulfuric acid

Molar Mass=98g/mol
Density=1.48g/ml
mass% =75%

- a) 20.2M
- b) 11.3M**
- c) 14.6
- d) 17.2M
- e) 0.0146

$$M = \frac{V \times d \times w\%}{MW}$$

$$M = \frac{1000 \times 1.48 \times 75\%}{98}$$

$$\underline{\underline{M = 11.3M}}$$

3) How many grams of HCl is needed to prepare 250ml of 3M solution of Hydrochloric acid :

Molar mass =36.5

Density = 1.2g/ml

Mass% = 37%

- a) 61.7g
- b) 74.0g**
- c) 22.8g
- d) 51.3g
- e) 70.1g

$$M = \frac{\text{mass}}{\frac{\text{molar mass}}{V(L)}}$$

$$\text{mass} = M \times V \times Mw$$

$$\text{mass} = 3 \times 0.25 \times 36.5 = 27.375$$

$$w\% = \frac{\text{mass solute}}{\text{mass solution}}$$

$$37\% = \frac{27.375}{\text{mass solution}}$$

$$\text{mass solution} = \underline{\underline{74 g}}$$

- 4) The weight of 95% $\text{Ca}_3(\text{PO}_4)_2$ (310.1767g/mol) required to prepare 500ml of 150 ppm of Ca^{+2} (40.078 g/mol) solution is :

- a) 0.129g
b) 0.203g
 c) 0.227g
 d) 0.193g
 e) 0.144g

$$\text{mol of } \text{Ca}_3(\text{PO}_4)_2 = 1/3 \text{ mol Ca} + 2$$

$$\frac{\text{mass} \times \text{wt}\%}{M.W} = 1/3 \frac{\text{conc} \times V}{M.W}$$

$$\frac{\text{mass} \times 95\%}{310.1767} = 1/3 \frac{0.150 \times 0.5}{40.078}$$

$$\text{Mass} = \underline{\underline{0.203g}}$$

- أهم شيء بالسؤال الوحدات, لازم الحجم يكون بال L , والوزنة بال gram
 ▪ $\text{Ppm} = \frac{\text{mg}}{\text{L}}$, $\text{mass (mg)} = \text{ppm} \left(\frac{\text{mg}}{\text{L}} \right) \times V(\text{L})$, $\text{g} = \text{mg} \times 10^{-3}$

- 5) The weight of 85% $\text{Ca}_3(\text{PO}_4)_2$ (310.1767g/mol) required to prepare 500ml of 150 ppm of PO^{-3} (95 g/mol) solution is :

- a) 0.227g
 b) 0.203g
c) 0.144g
 d) 0.129g
 e) 0.193g

$$\text{mol of } \text{Ca}_3(\text{PO}_4)_2 = 0.5 \text{ mol PO}_4$$

$$\frac{\text{mass} \times \text{wt}\%}{M.W} = 0.5 \frac{\text{conc} \times V}{M.W}$$

$$\frac{\text{mass} \times 0.85}{310.1767} = 0.5 \frac{0.150 \times 0.5}{95}$$

$$\text{Mass} = \underline{\underline{0.144 g}}$$

- 6) A sample of 0.1000g of NaCl (58.5g/mol) and 0.1155 g of MgCl_2 (95.3g/mol) were dissolved in 500 ml water, the part per million of Cl (35.5 g/mol) in the solution is

- a) 292**
 b) 214.0
 c) 150.0
 d) 236.0
 e) 269.1

$$1. \text{ Mass in NaCl} = \text{mass NaCl} \times \frac{MW \text{ Cl}}{MW \text{ NaCl}} = 0.1g \times \frac{35.5}{58.5} = 0.060 \text{ g}$$

$$2. \text{ Mass in MgCl}_2 = \text{mass MgCl}_2 \times \frac{2MW \text{ Cl}}{MW \text{ MgCl}_2} = 0.1155g \times \frac{71}{95.3} = 0.086 \text{ g}$$

$$3. \text{ mass Cl in sample} = \text{mass Cl in NaCl} + \text{in MgCl}_2 = 0.060 \text{ g} + 0.086 \text{ g} = 0.146 \text{ g}$$

$$4. \text{ Ppm} = \frac{\text{mg}}{\text{L}} = \frac{146\text{mg}}{0.5 \text{ L}} = 292 \text{ ppm}$$

7) The true statement of the following

is :

- 1) Random errors can be detected and corrected
- 2) The confidence level interval increases as the degree of confidence decreases
- 3) (How close together your data are) is defined as precision
- 4) Results that have lower coefficient of variation are more precise

- a) 1&2
- b) 2&3
- c) **3&4**
- d) 1&3
- e) 2&4

8) A sample of 0.1000 g of NaCl (58.5 g\mol) and 0.1155 g of CaCl₂ (111 g\mol) were dissolved in 500 ml water, the part per million of Cl⁻ (35.5 g\mol) in the solution is :

- a) 214.0
- b) 236.0
- c) 150.0
- d) 293.5
- e) **269.1**

$$1. \text{ Mass in NaCl} = \text{mass NaCl} \times \frac{M.W \text{ Cl}}{M.W \text{ NaCl}} = 0.1 \text{ g} \times \frac{35.5}{58.5} = 0.061 \text{ g}$$

$$2. \text{ Mass in CaCl}_2 = \text{mass CaCl}_2 \times \frac{2 M.W \text{ Cl}}{M.W \text{ CaCl}_2} = 0.1155 \text{ g} \times \frac{71}{111} = 0.07387 \text{ g}$$

$$3. \text{ Mass Cl in sample} = \text{mass Cl in NaCl} + \text{in CaCl}_2 = 0.061 \text{ g} + 0.07387 \text{ g} = 0.1345 \text{ g}$$

$$4. \text{ Ppm} = \frac{mg}{L} = \frac{134.55}{0.5 L} = \textbf{269.1}$$

9) A sample of 0.100 g of 90% Na₂SO₄ (142 g\mol) is dissolved in 500 ml of water, the volume of solution required to prepare 100 ml of 5.00 part per million Na⁺ (23 g\mol) is:

- a) **8.6 ml**
- b) 4.3 ml
- c) 6.4 ml
- d) 16.5 ml
- e) 3.2 ml

Degrees of freedom	3	4	5
t-value (95% level)	3.18	2.776	2.571

- 10) The results obtained by two analysts for a known standard samples of 48.60 ppm chloride are as follows:

Analyst 1: 49.32, 49.41, 49.66, 49.45

Analyst 2: 49.09, 49.08, 49.13, 49.10, 49.19, 49.45

The correct conclusion of the following is:

a) Both analyst are inaccurate

b) Both analyst are accurate

c) Analyst 1 is accurate, Analyst 2 is inaccurate

d) Analyst 1 is inaccurate, Analyst 2 is accurate

■ لأنه في الحالتين $T(\text{calc}) > T(\text{table})$

- 11) Suppose that 0.502 g of an unknown sulfate salt is dissolved in 50 mL of water. The solution is acidified with 6 M HCl, heated, and excess of aqueous BaCl₂ is slowly added to the mixture resulting in the formation of a white precipitate. If 0.433 g of precipitate is recovered calculate the percent by mass of SO₄²⁻ in the unknown salt. (molar mass of precipitate is 233.32 g/mol, and molar mass of SO₄²⁻ is 96 g/mol).

a) 46.0%

B) 55.2 %

C) 74.5 %

D) 35.5 %

12)

Degrees of freedom	3	4	5
t-value (95% level)	3.18	2.776	2.571

The following data are results of the determination of percent of HCl obtained by a student:

53.50, 53.00, 57.00, 52.00 and 52.50

The range you are 95% confident that the true value lies within is:

- a) 51.32 – 55.88
- b) 59.98 – 74.52
- c) 48.10 – 59.10
- d) **51.14 – 56.06**

13)

Dgrees of freedom	3	4	5
t- value (95 % level)	3.18	2.776	2.571

The following data are results of the determination of ppm of Na_2CO_3 obtained by a student and 105.0 , 104.0 , 106.0 , 107.0

The range you are 95% confident that the true value lies within is:

- a) 112.7 – 101.7
- b) 114.0 – 107.2
- c) **102.3 – 112.1**
- d) 110.9 – 103.5

Dgrees of freedom	3	4	5
t-value (95 % level)	3.18	2.776	2.572

The results obtained by two analysts for a known standard samples of 49.40 ppm chloride are as follows ;

Analyst 1: 49.32, 49.41, 49.66, 49.45

Analyst 2: 49.09, 49.08, 49.13, 49.10, 49.19, 49.45

The correct conclusion of the following is:

e) **Analyst 1 is accurate, analyst 2 is inaccurate**

f) Both analyst are inaccurate

g) Analyst 1 is inaccurate, Analyst 2 is accurate

h) Both analyst are accurate

15) Ingravimetric analysis of sulfate with barium, the product will be higher then expected

I) The precipitate is weighed without drying

II) The solution is filtered without heating

III) The precipitate is done in acidic medium

IV) The precipitate is dried without washing

V) The precipitating agent is rapidly added

Select one :

☐ III & V

☒ **I & IV**

☐ V

☐ II & V

☐ I

16) In the precipitation of barium sulfate, the particle size will be small when :

- I) It is precipitated from concentrated solution
- II) Precipitating BaCl_2 is added rapidly
- III) It is precipitated in neutral medium
- IV) It is precipitated from cold solution

Select one :

- ☐ I & III
- ☐ II & IV
- ☐ None of them
- ☒ **All of them**
- ☐ III & IV

17) A mixture of 100 mg of CaCl_2 (111 g/mol) and 150 mg of NaCl (58.45 g/mol) was dissolved and diluted to 500 ml, the part per million of chloride (35.45 g/mol) in the solution is :

- a) 155
- b) 246
- c) 310**
- d) 123

نفس فكرة سؤال 6 بس ركز هون معطيك الكتلة بالمل جرام حولها لجرام عن طريق الضرب ب 10^{-3}

The mass CaCl_2

18) Suppose that 0.200 g of an unknown sulfate salt is dissolved in 50 ml of water. The solution is acidified with 6 M HCl, heated, and an excess of aqueous BaCl₂ is added to the mixture resulting in the formation of a white precipitate. If 0.433 g of BaSO₄ precipitate is recovered, calculate the percent by mass of SO₃ in the unknown salt (molar mass of BaSO₄ is 233.32 g/mol, and molar mass of SO₃ is 80 g/mol) :

a) 74.2 %

b) 35.5 %

c) 46.0 %

d) 55.2 %

19) A sample of 0.1000 g of CaCl₂ (111 g/mol) and 0.1000 g of MgCl₂ (95.5 g/mol) were dissolved in 1000 ml water, the part per million of Cl⁻ (35.5g/mol) in the solution is :

a) 138.5

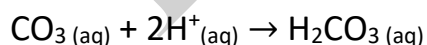
b) 276.9

c) 137.7

d) 68.85

e) 213.0

20) A sample of 0.9709 g sodium carbonate is dissolved in water and diluted to 100 ml. A sample of 25.00 ml is taken and titrated to the equivalence point with 16.90 ml of a 0.1022 M HCL solution. The percentage of Na₂CO₃ (106 g/mol) in the sample .



a) 75.42 %

b) 37.71 %

c) 18.8 %

d) 97.09 %

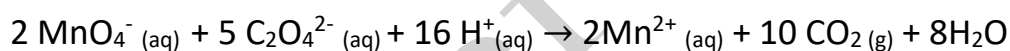
21) Four different objects were weighed by a person using two different balances ($t_{\text{tab}} = 3.182$)

	Object 1	Object 2	Object 3	Object 4
Balance 1	102.3	159.0	177.8	121.2
Balance 2	101.7	157.4	179.7	119.5

The correct statements of the following is :

- a) $T_{\text{cal}} > t_{\text{tab}}$ and there is no difference between balances
- b) $T_{\text{cal}} > t_{\text{tab}}$ and there is difference between balances
- c) $T_{\text{cal}} < t_{\text{tab}}$ and there is difference between balances
- d) $T_{\text{cal}} < t_{\text{tab}}$ and there is no difference between balances

22) A sample of 0.135 g of $\text{Na}_2\text{C}_2\text{O}_4$ (134 g/mol) was dissolved and titrated with KMnO_4 requiring 11.3 ml to reach the end point.



The molar concentration of KMnO_4 is :

- a) 0.0446
- b) 0.0357
- c) 0.111
- d) 0.0178